

Level 1 — 100 marks/question

L1-Q1. HCF (E)

C

CODE

```
#include <stdio.h>
int main()
{
    int n, a[100], i, hcf, lcm, x, y, temp;
    scanf("%d", &n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    hcf = lcm = a[0];
    for (i = 1; i < n; i++){
        x = hcf;
        y = a[i];
        while(y){
            temp = y;
            y = x % y;
            x = temp;
        }
        hcf = x;
        x = lcm;
        y = a[i];
        while(y){
            temp = y;
            y = x % y;
            x = temp;
        }
        lcm = (lcm / x) * a[i];
    }
    printf("HCF: %d\n", hcf);
    printf("LCM: %d", lcm);
    return 0;
}
```

INPUT

```
3
4 8 12
```

OUTPUT

```
HCF: 4
LCM: 24
```

L1-Q2. Armstrong (E)

C

CODE

```
#include <stdio.h>
int main()
{
    int lower, upper, i, temp, rem, digits, sum, first = 1;
    scanf("%d", &lower);
    scanf("%d", &upper);
    printf("Armstrong numbers between %d and %d are: ", lower, upper);
    for (i = lower; i ≤ upper; i++){
        temp = i;
        digits = 0;
        while (temp ≠ 0){
            digits++;
            temp /= 10;
        }
        temp = i;
        sum = 0;
        while (temp ≠ 0){
            rem = temp % 10;
            sum += pow(rem, digits);
            temp /= 10;
        }
        if (sum == i){
            if (first){
                printf("%d", i);
                first = 0;
            }
            else
                printf(", %d", i);
        }
    }
    return 0;
}
```

INPUT

100
500

OUTPUT

Amstrong numbers between 100 and 500 are: 153, 370, 371, 407

L1-Q3. series (E)

C

CODE

```
#include <stdio.h>
int main()
{
    int n, i, sum = 0;
    scanf("%d", &n);
    for (i = 1; i ≤ n; i++)
    {
        sum = sum + (i * i);
    }
    printf("Sum of series: %d", sum);
    return 0;
}
```

INPUT

5

OUTPUT

Sum of series: 55

L1-Q4. Number Pattern 2 (E)

C

CODE

```
#include <stdio.h>
int main()
{
    int i, j;
    for (i = 1; i ≤ 5; i++)
    {
        for (j=1; j≤i; j++)
        {
            printf("%d", j);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
1
12
123
1234
12345
```